Milestone 1: Team Engine Design

CSC 581 Game Engine Foundation

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Source Code: https://github.com/CSC581/shared-engine/tree/main

> *AI disclosure: Claude Code (Sonnet 5) was used to draft and re-format this content. All content was reviewed and edited by the team.*

Overview & Architecture

A reusable C++17/SDL3 foundation for 2D games: it owns the window, renderer, main loop, and timing, and gives games entities, physics, input, and collision.

Repository structure

```
shared-engine/
├── CMakeLists.txt       build for both libraries, the tests, and each game
├── include/             public headers (Engine, Game, Entity, Physics,
│                        Input, Collision)
├── src/                 engine implementation
├── tests/               unit tests for entities/physics, input, and collision
├── individual-games/    individual games built on the shared engine, one
│                        folder per team member
└── vendored/SDL/        SDL3 as a git submodule, built from source
```

What the engine is

The engine is a small foundation for building 2D games with SDL3. It is split into two CMake libraries:

- **`engine-geometry`**, SDL-free: `Entity`, `Physics`, `Collision`. Pure geometry and math, so it needs no window or renderer.
- **`engine`**, SDL-dependent: `Engine`, `Input`. Publicly links `engine-geometry` and `SDL3::SDL3`, so any game that links `engine` gets everything transitively.

A game does not subclass or modify the engine. Instead it implements the `Game` interface and hands an instance to `Engine::run()`. The engine owns the window, renderer, main loop, and timing; it knows nothing about players, enemies, or scores.

Design approach

The engine takes an **object-oriented** approach, in three specific senses:

- **Encapsulation.** `Engine` and `Entity` keep all state private and expose it only through member functions, so a game can never put an object into an inconsistent state by writing a field directly.
- **Polymorphism at one seam.** `Game` is an abstract base class of three pure virtual functions. `Engine::run()` holds a `Game&` and calls through the virtual function table, so it drives any game without knowing the concrete type. This is the single extension point, and the reason the engine never needs to change when a new game is added.
- **RAII.** `Engine`'s constructor acquires the SDL subsystem, window, and renderer, and its destructor releases them in reverse order. Copy and move are deleted, so ownership of those raw handles is unique and cleanup happens exactly once.

Tradeoff: this is a deliberate choice of inheritance over callbacks: `Engine::run(Game&)` could instead take three `std::function`s or free functions and avoid a class entirely. Inheritance wins here because the three functions are never called independently, they share `Game`'s implicit state (a game's own entities, score, flags) for free, where a callback signature would need to smuggle that state through captures or an opaque pointer. The cost is real: every game writes a class, overrides three pure virtuals, and a contributor new to C++ needs to understand virtual dispatch before touching gameplay code, more ceremony than a free-function or single-`main.cpp` version would need for a small game.

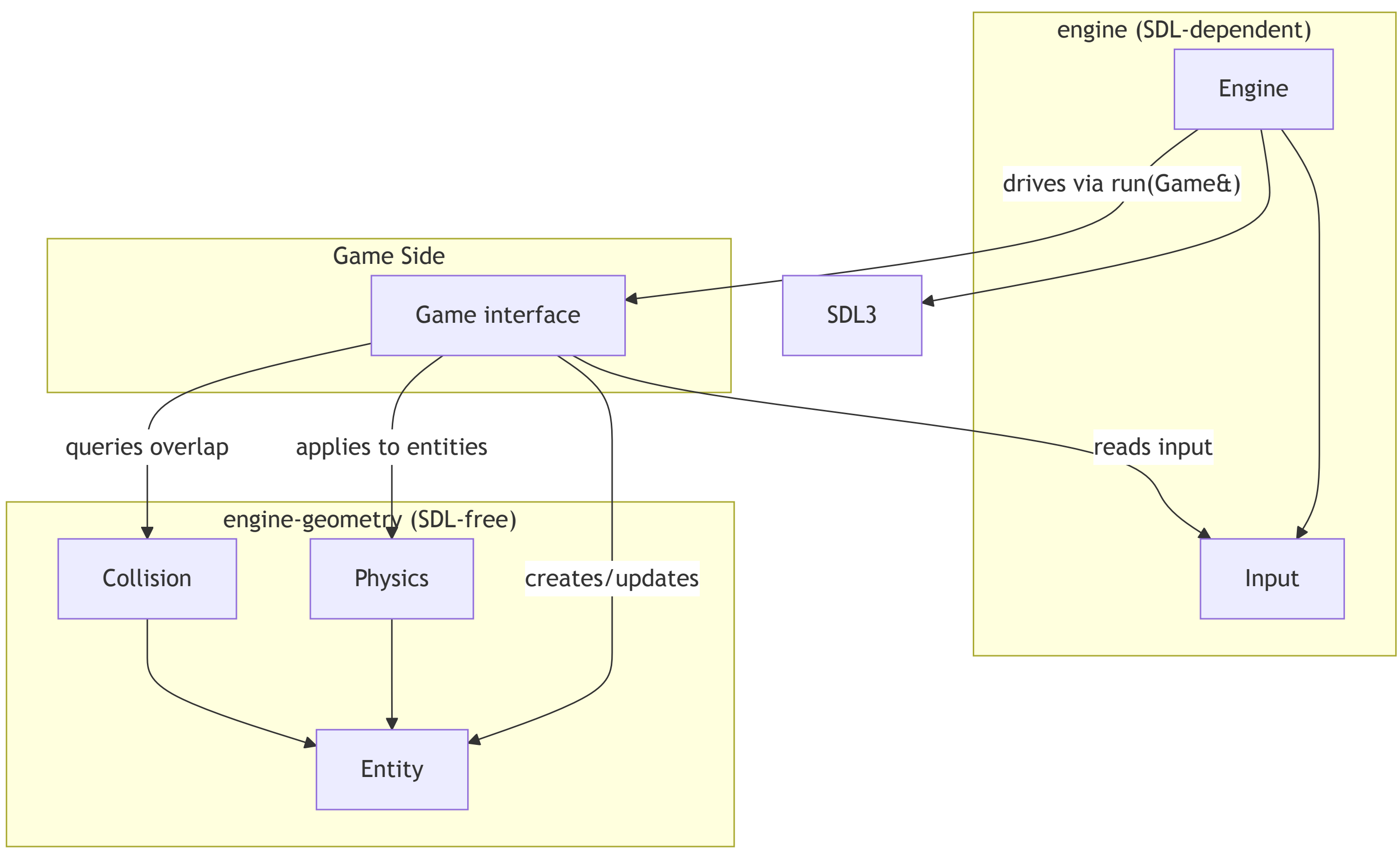
Not everything is modelled as an object, deliberately. `Physics`, `Collision`, and `Input` are all-`static` classes used as namespaces: they hold either no state (`Collision` is pure functions) or a single global (`Physics::gravity_`, `Input`'s key arrays), so there is nothing per-instance worth constructing. `Entity` is a concrete class with no virtuals rather than a base for a

`Player` / `Enemy` hierarchy, games compose it as a member instead of inheriting from it.

Subsystems

System	Files	Covers
Engine & Game	`Engine.hpp` , `Engine.cpp` , `Game.hpp`	Window/renderer setup, main loop, timing, render scaling, the game extension point
Entity	`Entity.hpp` , `Entity.cpp`	Position, size, velocity, bounding rectangle
Physics	`Physics.hpp` , `Physics.cpp`	Configurable gravity
Input	`Input.hpp` , `Input.cpp`	Keyboard state, just-pressed/just-released detection, multi-key queries
Collision	`Collision.hpp` , `Collision.cpp`	AABB overlap, intersection region, separation, overlap resolution

Architecture

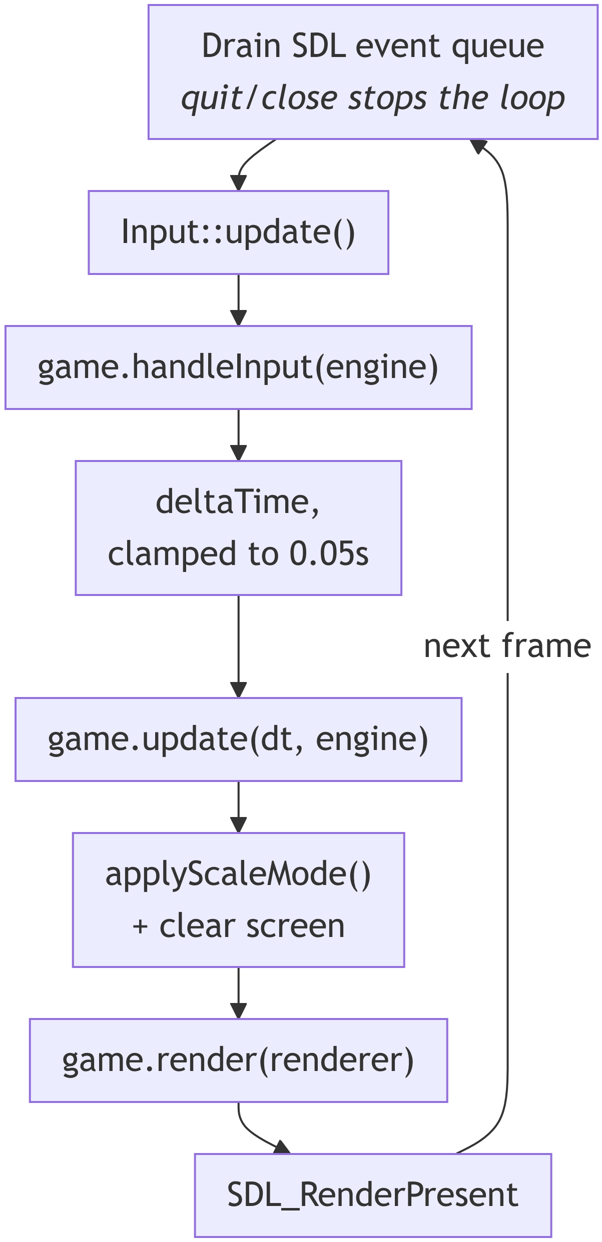


`Engine` never references `Entity`, `Physics`, or `Collision` directly, it only knows about `Game` and SDL. Everything below the `Game` line in the diagram is code a game chooses to use; the engine does not require it.

Tradeoff: keeping `engine-geometry` free of SDL is a deliberate split, not just a file layout. `Entity`, `Physics`, and `Collision` never include an SDL header, so they compile and run headless, `tests/CollisionTests.cpp` and the Entity/Physics tests check "do these rectangles overlap?" from the terminal, with no window, renderer, or `SDL_Init` involved. It also means this geometry survives a later switch from colored rectangles to sprites, since none of it depended on how anything is drawn; and it lines up with "Physics owns velocity, Collision owns bounds, Engine owns the window" as three independent concerns. The cost is two CMake targets instead of one: anything that genuinely needs both concerns (e.g. turning a mouse click's window-pixel coordinates into world-space geometry) has to live in `engine` and call down into `engine-geometry`, never the reverse, and a contributor adding a new file has to know which library it belongs in before an `#include <SDL3/SDL.h>` accidentally leaks into the SDL-free side.

Per-frame lifecycle

`Engine::run(Game& game)` drives one iteration of this sequence every frame, until the window is closed or `Engine::quit()` is called:



Why the order is what it is:

- The full event queue is drained even though the engine acts only on quit/close, that drain is what refreshes the array `SDL_GetKeyboardState()` reads. Every other event is discarded.
- `Input::update()` precedes both the scale-toggle check and `game.handleInput()`, so engine and game see identical key state.
- The frame timestamp is taken *after* `game.handleInput()`, so the measured interval covers the presented frame's own work.
- `applyScaleMode()` runs every frame, not just on resize, because it also resets the render scale to `1.0` before applying logical letterboxing.
- The engine clears before `game.render()` and presents after it returns, so `render()` only draws.

Build

From the repository root:

```
cmake -S . -B build
cmake --build build
```

Wire a game executable in `CMakeLists.txt` and link the `engine` target, which also puts `include/` on the include path:

```
add_executable(my-game individual-games/<name>/myGame.cpp)
target_link_libraries(my-game PRIVATE engine)
```

Code that needs only geometry can link `engine-geometry` alone, it is SDL-free and runs headless, with no window and no `SDL_Init`.

Engine & Game

`Engine.hpp` / `Engine.cpp`, window, renderer, main loop, timing, and render scaling. `Game.hpp`, the interface a game implements to plug into the engine.

Using it from a game

A game subclasses `Game`, implements the three callbacks, constructs an `Engine`, and hands the game to `Engine::run()`:

```
// After defining MyGame with the three Game callbacks:
try {
    Engine engine("My Game", 800, 600);
    engine.setClearColor(30, 60, 140);
    MyGame game;
    engine.run(game);
} catch (const std::exception& error) {
    std::cerr << "Engine failed to start: " << error.what() << '\n';
}
```

Because both constructor failure paths throw `std::runtime_error`, construction belongs inside a `try` / `catch` in `main()`.

`Engine`

`Engine` holds no game state of its own. It owns SDL setup and the frame loop, and drives whatever `Game` is passed to `run()`.

Construction and destruction

```
Engine(const char* title, int width, int height);
~Engine();
```

- `width` / `height` are the **design resolution**, the coordinate space a game lays its entities out in, not necessarily the window's pixel size (see Scale modes below).
- `SDL_Init(SDL_INIT_VIDEO)` runs first; on failure it throws `std::runtime_error` carrying `SDL_GetError()` 's message.
- `SDL_CreateWindowAndRenderer(...)` then creates an always-resizable window and its renderer together. If it fails, `SDL_Quit()` runs before throwing: the destructor never runs for an object whose constructor threw, so that call alone prevents a failed startup leaving SDL initialized.
- A private `applyScaleMode()` call finishes construction, so logical presentation matches the default `ScaleMode::Proportional` before the first frame.
- The destructor releases in reverse: renderer, window, `SDL_Quit()`.

Because `Engine` owns raw SDL handles with no reference-counting, all four copy/move operations are deleted. Games must draw with `getRenderer()` or the `SDL_Renderer*` passed to `Game::render`, and must never create a second renderer or call `SDL_Quit()` while an `Engine` is alive.

Running the game

```
void run(Game& game);
void quit();
```

- `run()` follows the lifecycle diagram above, blocking until `isRunning_` becomes `false`.
- `isRunning_` is cleared on `SDL_EVENT_QUIT`, `SDL_EVENT_WINDOW_CLOSE_REQUESTED`, or `engine.quit()`. `quit()` only flips the flag, the current frame still finishes before the loop exits.

- `deltaTime` is `(currentFrameTime - previousFrameTime) / 1000.0F` seconds from `SDL_GetTicks()`, capped at `0.05`. The cap keeps a stalled frame (window drag, breakpoint) from causing a large physics jump. There is no minimum clamp or fixed-timestep accumulator, the timestep is variable. `previousFrameTime` advances to the raw timestamp even when the delta is clamped, so time never accumulates a debt.

Games read keys through the polling `Input` API, never from `SDL_Event` directly. The clamp is a soft guard against a stalled frame teleporting a fast mover through a thin wall, not continuous collision detection.

Renderer and design-resolution accessors

```
SDL_Renderer* getRenderer() const;
int getWidth() const;
int getHeight() const;
```

- `getRenderer()` is the only supported way to obtain the renderer; games should not create their own.
- `getWidth()` / `getHeight()` return the design resolution passed to the constructor (constant for the `Engine`'s lifetime), not the current window pixel size. Games position entities in this coordinate space regardless of scale mode.

Clear color

`Engine::Color` holds `std::uint8_t` channels `red`, `green`, `blue`; default `{30, 60, 140}`, alpha always `255`.

```
void setClearColor(Color color);
void setClearColor(std::uint8_t red, std::uint8_t green, std::uint8_t blue);
```

The screen clears to `clearColor_` at the start of every frame's render step, before `game.render()` runs. `setClearColor` may be called any time (e.g. from `update()`) and takes effect starting with that same frame's clear.

Scale modes

```
ScaleMode getScaleMode() const;
void setScaleMode(ScaleMode mode);
void toggleScaleMode();
void setScaleToggleKey(SDL_Scancode key);
```

Games always draw in design-resolution coordinates (`getWidth() x getHeight()`). `applyScaleMode()` (private; called once from the constructor and once per frame from `run()`) is the single place that maps those coordinates onto window pixels.

Mode	Behavior
`Constant`	`SDL_LOGICAL_PRESENTATION_DISABLED` . One design unit maps to one screen pixel, regardless of window size. Resizing reveals more or less of the world rather than scaling it.
`Proportional` (default)	Logical presentation with `SDL_LOGICAL_PRESENTATION_LETTERBOX` at the design width/height. Uniform scale preserving the design aspect ratio; unused space is letterboxed rather than stretched.

- `setScaleMode()` sets the mode directly; `toggleScaleMode()` flips between the two. Both only touch the field and never talk to SDL, so the change takes effect on the next presented frame regardless of caller.
- `scaleToggleKey_` defaults to `SDL_SCANCODE_F1` . Each frame, if the key isn't `SDL_SCANCODE_UNKNOWN` and `Input::isKeyJustPressed(scaleToggleKey_)` is true, `run()` calls `toggleScaleMode()` itself, no wiring needed. This check runs before `handleInput` , so a game reading the same key sees the mode already in effect for the frame. `setScaleToggleKey(SDL_SCANCODE_UNKNOWN)` disables the built-in toggle; any other scancode rebinds it.

`applyScaleMode()` resets render scale to `(1, 1)` first so the two mechanisms can never compound, any prior mode's or game's leftover scale is wiped before the new presentation is set. `Proportional` relies on letterboxing rather than separate horizontal and vertical scale factors, which would stretch the image and distort the design aspect ratio.

Private state

All state is private; games reach it only through the public functions above.

Variable	Type	Purpose
`maxDeltaTime`	`static constexpr float` (= `0.05F`)	Upper bound on a frame's delta time, in seconds.
`defaultClearColor`	`static constexpr Color` (= `{30, 60, 140}`)	Clear colour a new engine starts with.
`window_`	`SDL_Window*`	The window, destroyed by the destructor.
`renderer_`	`SDL_Renderer*`	The one renderer; handed to `Game::render` each frame.
`width_` , `height_`	`int`	Design resolution, fixed at construction.
`clearColor_`	`Color`	Colour the loop clears to each frame.
`scaleMode_`	`ScaleMode`	Design-units-to-pixels mapping; `Proportional` by default.
`scaleToggleKey_`	`SDL_Scancode`	Built-in toggle key; F1 by default, `SDL_SCANCODE_UNKNOWN` disables it.
`isRunning_`	`bool`	Loop condition; cleared by `quit()` or a close event.

`Game`

```
virtual void handleInput(Engine& engine) = 0;
virtual void update(float deltaTime, Engine& engine) = 0;
virtual void render(SDL_Renderer* renderer) const = 0;
```

`Game` is the engine's only extension point: a game subclasses it and passes an instance to `Engine::run()` . `Engine` calls exactly one of each method per frame, in this order:

Method	When	Responsibility
`handleInput`	After `Input::update()` , so every `Input::` query reflects this frame	Read keys, set intent: velocity, jump flags, `engine.setScaleMode()` , `engine.quit()` .
`update`	After delta time is computed and clamped	Advance simulation: `Physics::applyGravity` , `Entity::update` , `Collision::resolve` , camera.
`render`	After clear and scale apply	Draw only. The engine calls `SDL_RenderPresent` when this returns, so `render()` must not clear or present. It is `const` , so it should not mutate game state.

`deltaTime` is in **seconds** and already clamped by the loop; games should not re-clamp it. `Game` has a virtual destructor and no other state, a game need not store an `Engine` reference beyond what's passed into each call.

A typical per-frame order inside a game's own code (not enforced by the engine):

Entity

`Entity.hpp` / `Entity.cpp` , the generic game object: position, size, and velocity, plus a bounding rectangle. It has no knowledge of rendering, input, or any specific game; it is pure state and simple motion.

Using it from a game

```
Entity player(100.0F, 200.0F, 32.0F, 32.0F);
player.setVelocity(200.0F, 0.0F);

// During the game's update:
player.update(deltaTime);
```

Each Entity starts with zero velocity, so it stays still until a setter or gravity changes it. For falling entities, call `Physics::applyGravity(player, deltaTime)` before `player.update(deltaTime)`. Both calls are explicit, the engine never applies gravity or moves Entities on its own, so a stationary platform can skip both and a flying object can move without gravity.

`Rect`

A plain axis-aligned rectangle: `x` / `y` is the top-left corner, `width` / `height` extend right and down from it. `Rect` is a value type with no behavior of its own, the Collision section supplies the geometry operations on it.

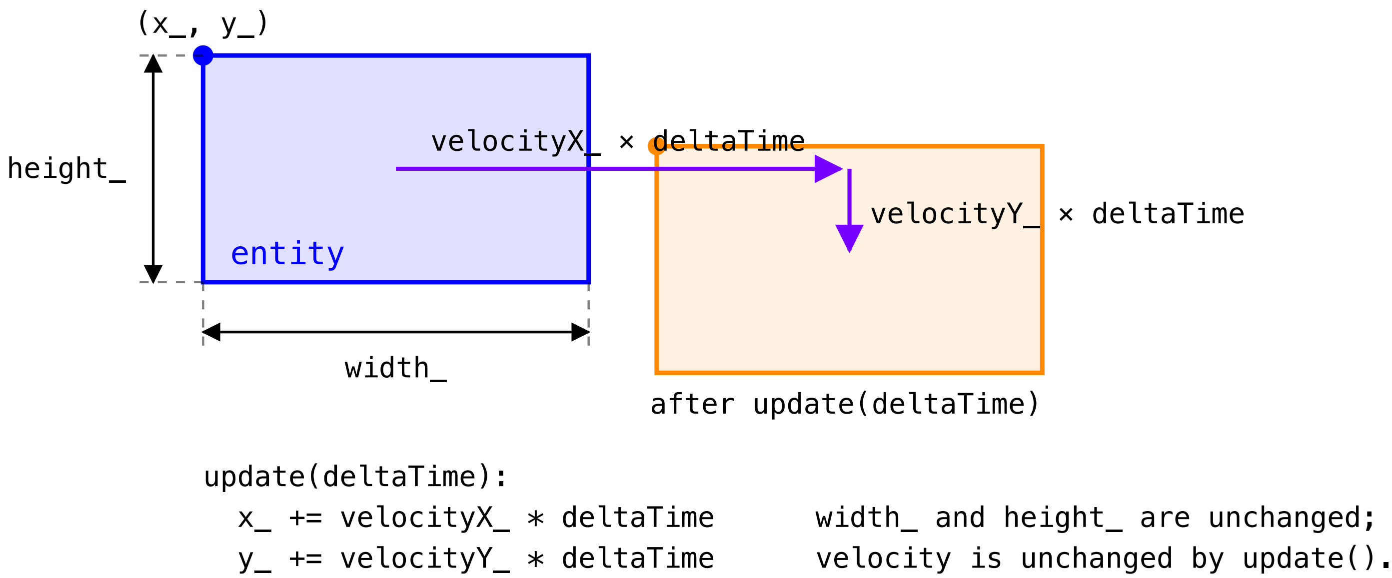
Construction and state

```
Entity(float x, float y, float width, float height);
```

An `Entity` stores position (`x_` , `y_`), size (`width_` , `height_`), and velocity (`velocityX_` , `velocityY_`). Velocity is default-initialized to `0.0F` in the class definition and not touched by the constructor, so a newly created entity is stationary until a velocity setter is called.

Variable	Type	Purpose
`x_` , `y_`	`float`	Top-left position of this Entity.
`width_` , `height_`	`float`	Width and height of this Entity's rectangle.
`velocityX_` , `velocityY_`	`float`	Movement in units per second; both start at `0.0F` .

All state is private. Each Entity owns its own copy, so changing one Entity's velocity never affects another.



Motion

```
void update(float deltaTime);
```

`update()` performs simple explicit (forward) Euler integration of velocity into position, each axis advances by that axis's velocity times `deltaTime` :

```
x += velocityX * deltaTime
y += velocityY * deltaTime
```

It does not touch velocity itself: anything that changes velocity over time (gravity, acceleration, drag) is a separate step the game performs before calling `update()` , most commonly via `Physics::applyGravity` . There is no maximum-speed clamp, drag, or friction built in.

`deltaTime` is in seconds and velocity is in units per second, so a horizontal velocity of `200.0F` moves the Entity 100 units in half a second. Scaling by elapsed time keeps movement consistent across frame rates.

Setters and getters

```
void setPosition(float x, float y);
void setSize(float width, float height);
void setVelocity(float velocityX, float velocityY);
void setVelocityX(float velocityX);
void setVelocityY(float velocityY);
float getX() const;
float getY() const;
float getVelocityX() const;
float getVelocityY() const;
Rect getBounds() const;
```

- `setPosition` / `setSize` overwrite both axes/dimensions at once. `setPosition` leaves velocity intact, so a moving Entity keeps moving from its new position on the next `update()` ; use it to place, teleport, or reset. No setter moves the Entity, position changes only in `update()` .
- `setVelocityX` / `setVelocityY` change one axis without disturbing the other, used for example, when `Collision::resolve` zeroes velocity on only the axis it pushed along.
- `getBounds()` returns a fresh `Rect{x_, y_, width_, height_}` **by value**, computed on every call rather than cached. It always reflects current position and size, but the returned `Rect` is disconnected from the Entity: editing it has no effect, and it does not track later movement.
- There are no direct getters for size; read `width_` / `height_` via `getBounds()` .

Collision convenience methods

```
bool collidesWith(const Entity& other) const;
bool containsPoint(float x, float y) const;
```

These are declared on `Entity` (in `Entity.hpp`) but **implemented in `Collision.cpp`** , not `Entity.cpp` . They delegate to `Collision::intersects(*this, other)` and `Collision::contains(getBounds(), x, y)` .

This split is intentional: it lets a game write the natural `player.collidesWith(wall)` call on `Entity` itself, while keeping

`Entity.cpp` / `Entity.hpp` free of any dependency on the collision system. `Entity` depends only on its own header; `Collision.cpp` depends on both.

Physics

`Physics.hpp` / `Physics.cpp` : a single configurable value (gravity) and one function that applies it to an entity. It depends only on `Entity` , not on `Engine` or SDL.

API

```
static void setGravity(float gravity);
static float getGravity();
static void applyGravity(Entity& entity, float deltaTime);
```

Everything on `Physics` is static: the class is effectively a namespace holding one piece of shared, global data: `gravity_` .

Variable	Type	Purpose
`gravity_`	`static float`	Acceleration used by every `applyGravity()` call; defaults to `980.0F` , in units per second squared.

- `980.0F` corresponds to roughly Earth gravity when a design unit is treated as a centimeter (or as an arbitrary "pixels per second squared" value a game is free to reinterpret).
- `setGravity()` overwrites `gravity_` for all subsequent `applyGravity` calls, on any entity, there is no per-entity gravity scale.
- `applyGravity(entity, deltaTime)` only **adds to vertical velocity**; it does not move the entity. Gravity accumulates onto whatever `velocityY` already is, so calling it every frame produces an accelerating fall; position changes only when `update(deltaTime)` runs afterward. With gravity at `980.0F` , a call with `deltaTime == 0.5F` adds `490.0F` to vertical velocity; horizontal velocity and position are unchanged.
- Positive gravity accelerates in the positive Y direction; negative gravity pushes entities upward instead.
- Setting gravity to `0.0F` or skipping `applyGravity` for a frame stops it *adding* velocity but does **not** clear velocity an Entity already has;

an entity already falling keeps falling at its current speed.

Usage pattern

Gravity is opt-in per entity and per frame: nothing calls it automatically. A game decides which entities should fall and calls it from its own `update()` :

```
Physics::setGravity(980.0F);

// each frame, for entities that should fall:
Physics::applyGravity(player, deltaTime);
player.update(deltaTime);
```

Order matters: gravity changes velocity first, then movement uses that new velocity the same frame. Reversing the calls makes an entity move on the previous frame's velocity.

Because `gravity_` is a single static value, giving different entities different gravity simultaneously (e.g. a slow-motion pickup affecting only one entity) requires scaling the effect manually by calling `setGravity` with a different value before applying it and restoring it afterward, or by skipping `applyGravity` for that entity and adding vertical velocity by hand.

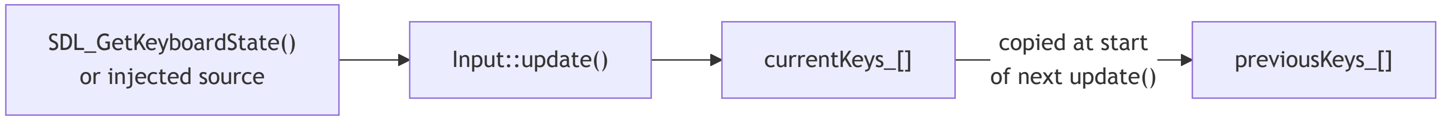
Input

`Input.hpp` / `Input.cpp` : keyboard state for games built on the engine.

`Input` is part of `engine` (SDL-dependent), not `engine-geometry` , because it reads `SDL_GetKeyboardState` from SDL. Everything on it is static (global) state, similar in spirit to `Physics` .

Design: polling only

Games query key state by polling (`Input::isKeyPressed(...)`). `Input` has no event-queue path of its own: one read of the keyboard state per frame is the single source of truth for every query.



`Engine::run()` drains the SDL event queue itself (acting only on quit/close), then calls `Input::update()` once per frame before `game.handleInput()` runs. Game code normally calls only the query methods below; `update()` is engine-driven.

Tradeoff: a press *and* release that both fall between two `update()` calls is invisible to the engine. Latching key-down events from the SDL queue would close that gap, but at the cost of a second source of truth that can report a key as held when the hardware already says it is not. Polling only keeps every query answering one unambiguous question, "is this key down right now?", which is what continuous movement and held-modifier chords are built on.

Using it from a game

```
// Inside Game::handleInput():
const float dx = Input::getAxis(SDL_SCANCODE_A, SDL_SCANCODE_D);
const float dy = Input::getAxis(SDL_SCANCODE_W, SDL_SCANCODE_S);
player_.setVelocity(dx * speed_, dy * speed_);

if (Input::isKeyJustPressed(SDL_SCANCODE_SPACE)) {
    jump();
}
```

State arrays

`keyCount = SDL_SCANCODE_COUNT`: every SDL scancode has its own slot, so any number of keys can read as held simultaneously (diagonal movement, run + jump, modifier chords, two players on one keyboard) with no special-casing. `isValidKey` guards every public lookup so an out-of-range `SDL_Scancode` cannot read or write outside the arrays.

Variable	Type	Purpose
`keyCount`	`static constexpr int` (= `SDL_SCANCODE_COUNT`)	Size of every key array, and the valid scancode range.

Variable	Type	Purpose
`currentKeys_`	`std::array<bool, keyCount>`	Keys held during the current frame.
`previousKeys_`	`std::array<bool, keyCount>`	Last frame's `currentKeys_` .
`stateSource_`	`const bool*`	Injected keyboard array, or `nullptr` to read real hardware.
`stateSourceKeyCount_`	`int`	Entries in `stateSource_` ; `0` when there is no injected source.

All state is private and static. Both arrays are indexed directly by `SDL_Scancode` , so `currentKeys_[SDL_SCANCODE_W]` is the state of the W key.

`Input::update()`

Each call:

1. Copies `currentKeys_` into `previousKeys_` , so this frame's state becomes "last frame" for just-pressed/just-released detection. 2. Resets `currentKeys_` to all- `false` , then reads the active keyboard-state source into it. Only `min(availableKeys, keyCount)` entries are copied, so a shorter source array cannot overrun `currentKeys_` ; clearing first keeps a key that disappears from the source from sticking.

Both presses and releases are detected purely by comparing `currentKeys_` against `previousKeys_` , so the whole edge-detection story lives in this one function.

Query API

```
static bool isKeyPressed(SDL_Scancode key);
static bool isKeyJustPressed(SDL_Scancode key);
static bool isKeyJustReleased(SDL_Scancode key);
```

- `isKeyPressed(key)` , `currentKeys_[key]` is true this frame (held, regardless of how long).

- `isKeyJustPressed(key)` , true this frame, false last frame. Fires for exactly one frame per press.
- `isKeyJustReleased(key)` , false this frame, true last frame. Fires for exactly one frame per release.
- All three return `false` for an out-of-range scancode.

Multi-key helpers

```
static bool areAllKeysPressed(std::initializer_list<SDL_Scancode> keys);
static bool isAnyKeyPressed(std::initializer_list<SDL_Scancode> keys);
static int pressedKeyCount();
static std::vector<SDL_Scancode> getPressedKeys();
static float getAxis(SDL_Scancode negativeKey, SDL_Scancode positiveKey);
```

- `areAllKeysPressed({...})` : every listed key is held; useful for chords and modifier combos. An **empty** list returns `false` , not vacuously `true` : an explicit check is needed because `std::all_of` over an empty range is `true` , which would make an empty chord fire every frame. `isAnyKeyPressed({...})` needs no such guard, since `std::any_of` already returns `false` for an empty range.
- `pressedKeyCount()` : count of scancodes currently pressed, across the whole keyboard.
- `getPressedKeys()` : currently-pressed scancodes, ascending order. Allocates a `std::vector` per call, so it belongs in rebinding screens, debug overlays, and logging rather than a hot per-frame path.
- `getAxis(negativeKey, positiveKey)` : `-1.0F` if only `negativeKey` is held, `+1.0F` if only `positiveKey` is held, `0.0F` if neither or **both** are. Uses two independent `if` statements rather than `if` / `else` , so opposing keys cancel instead of one silently winning. Calling it twice with two axis pairs (e.g. A/D and W/S) gives a full 8-way direction with no extra bookkeeping.

Keyboard state source

```
static void setKeyboardStateSource(const bool* keyboardState, int keyCount);
```

By default `update()` reads from `SDL_GetKeyboardState` . Calling `setKeyboardStateSource(myArray.data(), myArray.size())` swaps that for an arbitrary `bool` array, so a caller can supply keyboard state without any real SDL event loop or window. Passing `nullptr` restores real hardware reads. This is the only supported way to feed `Input` a synthetic state.

The injected array is **not copied**, only the pointer and length are stored, so it must outlive every `update()` call that reads it. The length is reset to `0` whenever the pointer is null, so a stale length can never be paired with a null pointer.

Collision

`Collision.hpp` / `Collision.cpp` , axis-aligned bounding-box (AABB) collision geometry.

`Collision` only answers geometric questions, "are these two boxes overlapping, and by how much?", and never decides what a game should do in response. Deciding what a collision *means* (stop the player, take damage, trigger a checkpoint) is entirely up to the game.

Tradeoff: this division is deliberate, but the boundary isn't perfectly clean. `getSeparation` / `resolve` do make one decision on the game's behalf: which axis to push along. They compare the overlap's width against its height and push along whichever is **smaller**, the shallower penetration, on the theory that it is the minimal correction (see Separation and resolution below). That is a purely geometric heuristic, though, not a game-aware one: `Collision` still doesn't know that a push along Y means "the player landed on a platform" or that a push along X means "a bullet grazed a wall." The same `intersects` / `resolve` calls work whether a hit should stop a player, cost a life, award a coin, or destroy a block, but the engine has no vocabulary for any of those outcomes, there is no `isGameOver` , `isCollectible` , or `isGrounded` anywhere in `Collision` or `Entity` . Every game reads the resolved push axis/direction (or the raw AABB overlap) and re-derives those concepts itself in its own collision-response code.

All methods are `static` and `Collision` holds no state. Because every entry point is a pure function of its arguments, the same inputs always give the same answer, and nothing has to be initialised or reset between frames.

Using it from a game

Include `Collision.hpp` and call the static functions, there is no `Collision` instance to construct. This excerpt assumes `player` and `coin` are Entities and `button` is a Rect:

```
if (Collision::intersects(player, coin)) {
    collectCoin();
}

const Rect overlap = Collision::getIntersection(player, coin);

if (Collision::contains(button, mouseX, mouseY)) {
    press();
}
```

`Entity` also exposes two conveniences that read more naturally at the call site `player.collidesWith(wall)` and `player.containsPoint(mouseX, mouseY)` declared in `Entity.hpp` but implemented here; see the Entity section for why.

Where it goes in the frame

Inside `Game::update()` , after the engine has already run `handleInput()` :

```
Physics::applyGravity(...)   // optional, per entity
entity.update(deltaTime)     // integrate velocity into position
handleCollisions()           // detect / separate / resolve
// Game::render() then draws the resolved positions
```

Resolve **after** movement, so the overlap reflects this frame's integration. The engine's `deltaTime` clamp (`Engine::maxDeltaTime` , 0.05 s) limits how far an entity can travel in one step, but it does not replace careful collision logic for thin surfaces.

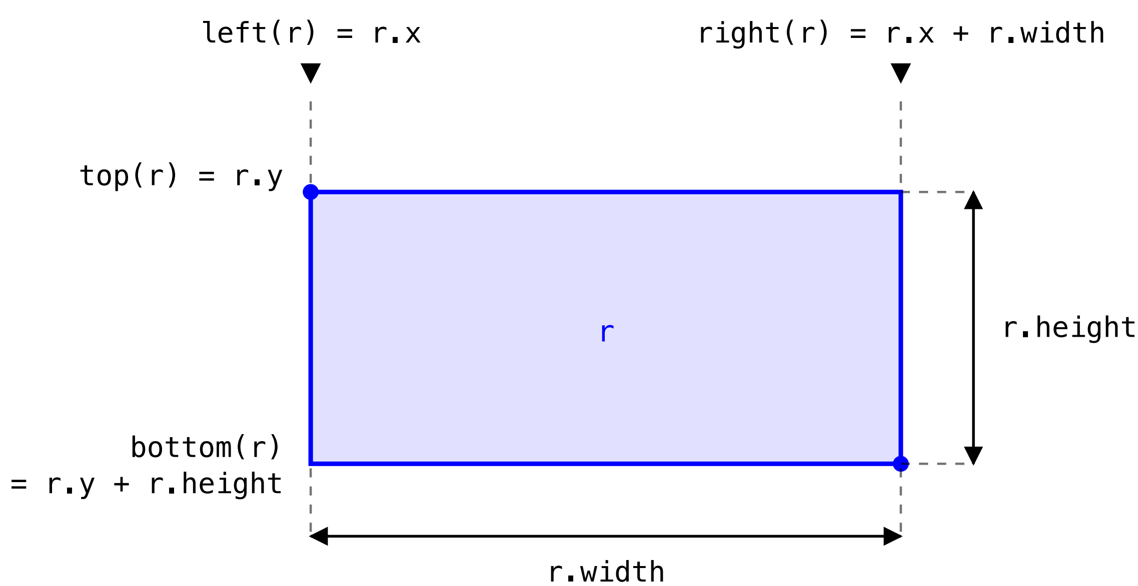
Geometry conventions

`Collision.cpp` defines four unexported helpers, `left(rect)` (`rect.x`), `right(rect)` (`rect.x + rect.width`), `top(rect)` (`rect.y`), and `bottom(rect)` (`rect.y + rect.height`). Everything else in the file is written in terms of these, so the edge convention lives in exactly one place.

The geometry they work on is `Rect` , declared in `Entity.hpp` , with the origin at the box's **top-left**:

Field	Type	Meaning
`x`	`float`	Left edge, in world units.
`y`	`float`	Top edge (y grows downward, as in SDL).
`width`	`float`	Extent to the right of `x` .
`height`	`float`	Extent below `y` .

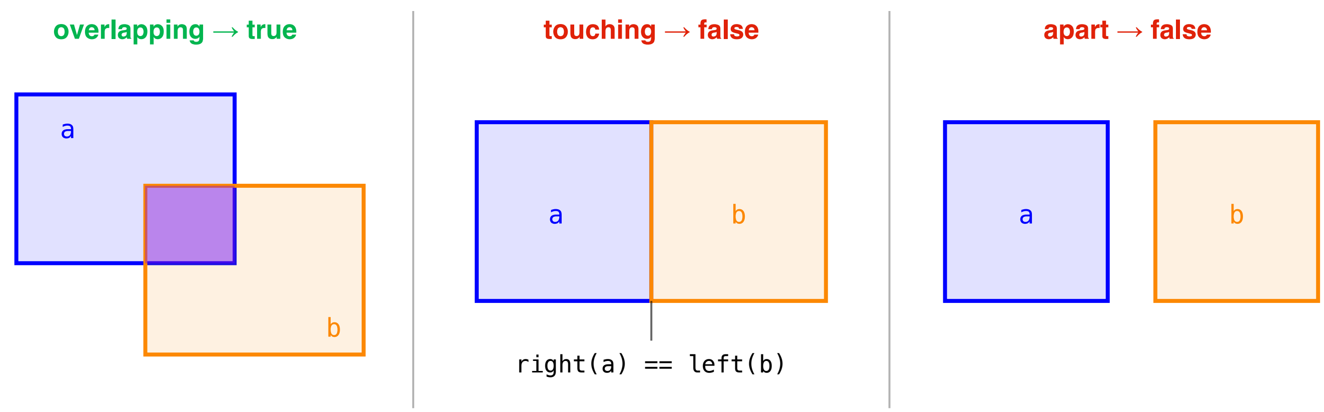
Because y increases downward, moving "up" climbing or jumping means *decreasing* `y` and a *negative* `velocityY` . `Entity::getBounds()` builds a `Rect` from an entity's position and size.



Overlap detection

```
static bool intersects(const Rect& a, const Rect& b);
static bool intersects(const Entity& a, const Entity& b);
```

- **Touching edges do not count as a collision**: the comparisons are strict (`<` / `>` , not `<=` / `>=`), so two boxes that share exactly one edge (e.g. one ends at `x=100` and the other starts at `x=100`) are reported as not intersecting. That is what lets a game slide an entity flush against a wall without the wall reporting a hit every frame.
- **A zero-size box never collides with anything**, including itself, guarded explicitly before the geometric check.
- **An entity does overlap itself** (same box, positive size), and detection is symmetric: `intersects(a, b) == intersects(b, a)` .
- **A fully-contained box does count as intersecting** its container.
- The `Entity` overload simply calls the `Rect` overload on `a.getBounds()` / `b.getBounds()` .

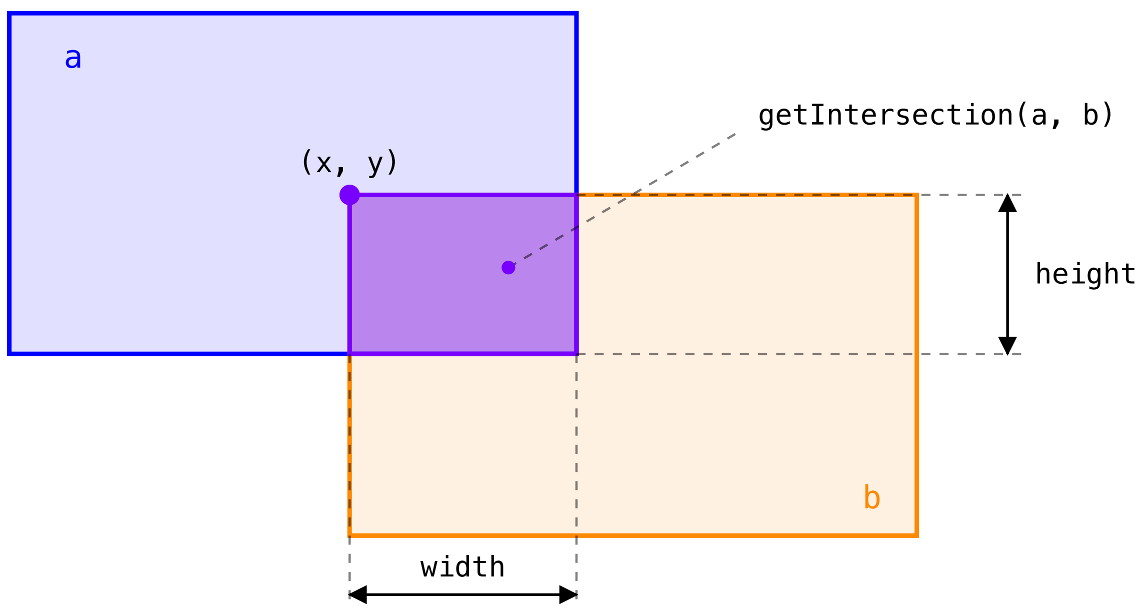


Intersection region

```
static Rect getIntersection(const Rect& a, const Rect& b);
static Rect getIntersection(const Entity& a, const Entity& b);
```

Returns the overlapping rectangle: `{0, 0, 0, 0}` if the boxes do not intersect per `intersects()` above, which also means touching-only boxes report a zero-size intersection, not a zero-width sliver at the shared edge. When they do overlap, the result spans the larger left/top edges (`x = max(left(a), left(b))` , `y = max(top(a), top(b))`) to the smaller right/bottom ones (`width = min(right) − x` , `height = min(bottom) − y`), useful for a game that wants to react to *how deep* an overlap is, not just whether one exists.

Past the guard both extents are guaranteed positive, because the boxes are known to overlap strictly. So a zero width and height always mean "no intersection", and depth-based reactions need no separate check.

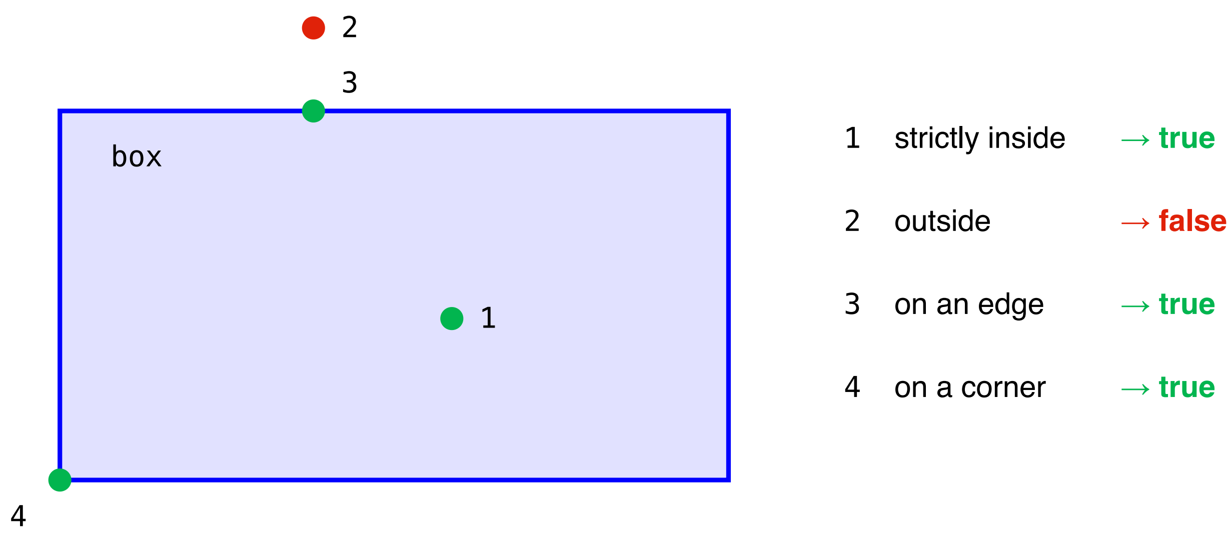



```
x      = max(left(a),   left(b))
y      = max(top(a),    top(b))
width  = min(right(a),  right(b))  - x
height = min(bottom(a), bottom(b)) - y
```

Point containment

```
static bool contains(const Rect& box, float x, float y);
```

Inclusive on all four edges, unlike `intersects`, which is exclusive, a point exactly on the boundary or a corner counts as contained. A click on a button's border should press it.



Separation and resolution

`getSeparation`

```
static bool getSeparation(const Entity& moving, const Entity& blocker,
                          float& outX, float& outY);
```

Computes the smallest push, along a single axis, that would separate `moving` from `blocker`:

1. Compute the overlap rectangle between `moving` and `blocker` via `getIntersection`. If either dimension is `<= 0`, there is no collision: return `false` and leave `outX` / `outY` untouched.
2. Pick the axis with the **smaller overlap dimension**, that is the shallowest penetration, and pushing along it is the minimal correction:
   - If `overlap.width < overlap.height`: push along X. Direction is `-1` if `moving`'s horizontal center is left of `blocker`'s center, else `+1`. `outX = overlap.width * direction`, `outY = 0`.
   - Otherwise: push along Y, using the same center comparison on the vertical axis, with `outY = overlap.height * direction`, `outX = 0`.
3. Return `true` and write `outX` / `outY`.

Only one axis is ever pushed, this is a single-axis minimum-translation resolution, not a full 2D minimum-translation-vector solve. When the overlap is exactly square (`width == height`), the `<` comparison means the **vertical** axis is chosen (ties go to Y/height). The center comparison uses `left(a) + right(a)` : a doubled midpoint, since the factor of two cancels and no division is needed.

`getSeparation()` reports the push instead of applying it, so a game can inspect the direction before deciding what to do:

```
float pushX = 0.0F;
float pushY = 0.0F;

if (Collision::getSeparation(player, platform, pushX, pushY)) {
    if (pushY < 0.0F) {
        // Pushed upward, so treat this as a landing candidate.
    }
}
```

`resolve`

```
static bool resolve(Entity& moving, const Entity& blocker);
```

Convenience response built directly on `getSeparation` . It moves `moving` out of `blocker` by the separation vector and zeroes velocity **only on the axis that was pushed**, resolving a vertical landing zeroes `velocityY` (stops the fall) but leaves `velocityX` untouched (horizontal movement survives landing on the ground), and vice versa for a horizontal wall push. It returns `false` and leaves `moving` unmodified if the two were not overlapping.

Worked examples

Detect only. The game supplies the whole response:

```
if (player.collidesWith(hazard)) {
    player.setPosition(spawnX, spawnY);
    player.setVelocity(0.0F, 0.0F);
}
```

Stop against a wall. A shallow horizontal overlap pushes sideways and clears `velocityX` only:

```cpp
Entity player(90.0F, 0.0F, 100.0F, 100.0F);
player.setVelocity(500.0F, 200.0F);
Entity wall(180.0F, 0.0F, 100.0F, 100.0F);

Collision::resolve(player, wall);
// player.x == 80, velocityX == 0, velocityY still 200
```

Land on the ground. A shallow vertical overlap pushes up and clears `velocityY` only:

```cpp
Entity faller(0.0F, 95.0F, 100.0F, 100.0F);
faller.setVelocity(50.0F, 300.0F);
Entity ground(0.0F, 180.0F, 400.0F, 100.0F);

Collision::resolve(faller, ground);
// faller.y == 80, velocityY == 0, velocityX still 50
```